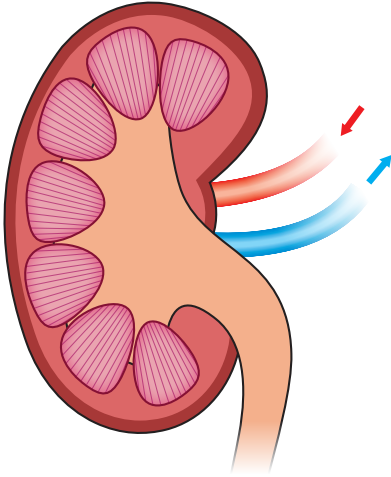


Answer **all** questions.

1. **Image 1.1** shows a section through a human kidney.

Image 1.1



- (a) **Draw arrows on Image 1.1** to label the following parts: [2]
- (i) the medulla;
 - (ii) the renal artery.

(b) **Table 1.2** shows the results of an investigation on a patient in a hospital.

Table 1.2

Substance	Concentration in blood entering kidney (au)	Concentration in blood leaving kidney (au)	Present in urine (✓ or ×)
glucose	145	32	✓
protein	120	120	
urea	93	0	

- (i) I. **Complete Table 1.2** to show which substance(s) would be present in the urine of this patient. One row has been done for you. [1]
- II. State which **one** of the substances is a waste product. [1]
-
- III. Doctors suggested that this patient had diabetes. From **Table 1.2**, state the evidence to support this suggestion. [1]
-
-



Examiner
only

(ii) Describe the chemical tests which could be carried out to confirm whether or not the urine contained glucose and protein. For each test state the colours you would expect **for the results shown in Table 1.2.**

I. glucose test [3]

.....

.....

.....

II. protein test [2]

.....

.....

(iii) **Complete the risk assessment** below which shows one hazard linked to these tests. [1]

Hazard	Risk	Control measure
Chemical reagent is an irritant		
		
		
		

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03

